



Recent Hypergraph Contrastive Learning

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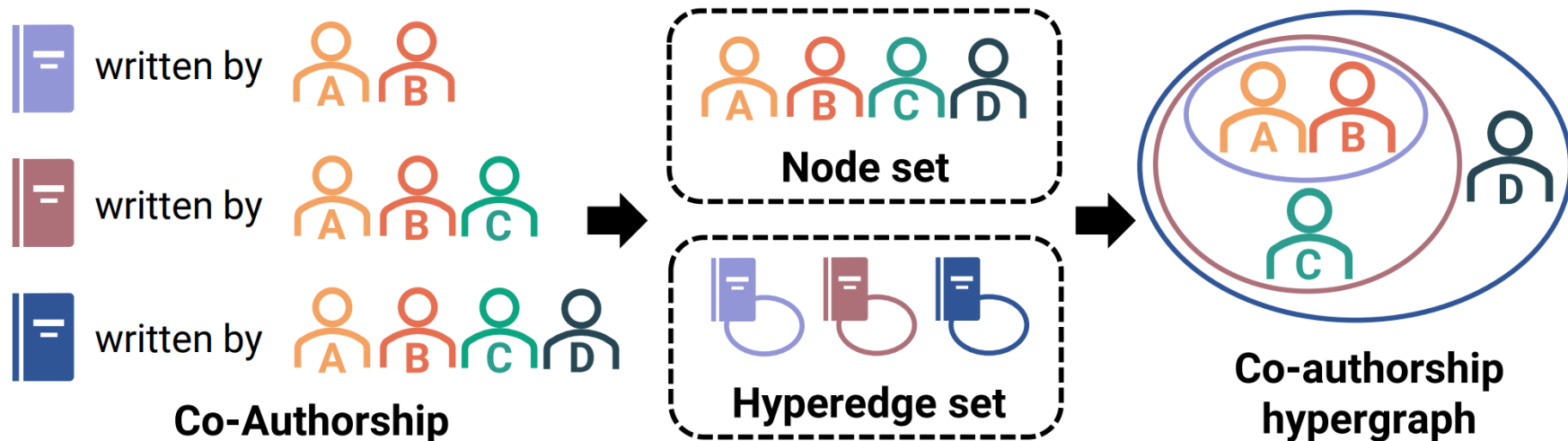
MINDS Lab

- ☐ Introduction
- ☐ CHGNN: A Semi-Supervised Contrastive Hypergraph Learning Network
- ☐ Heterophily-aware Contrastive Learning for Heterophilic Hypergraphs
- ☐ Conclusion

Hypergraphs Model Higher-Order Interactions

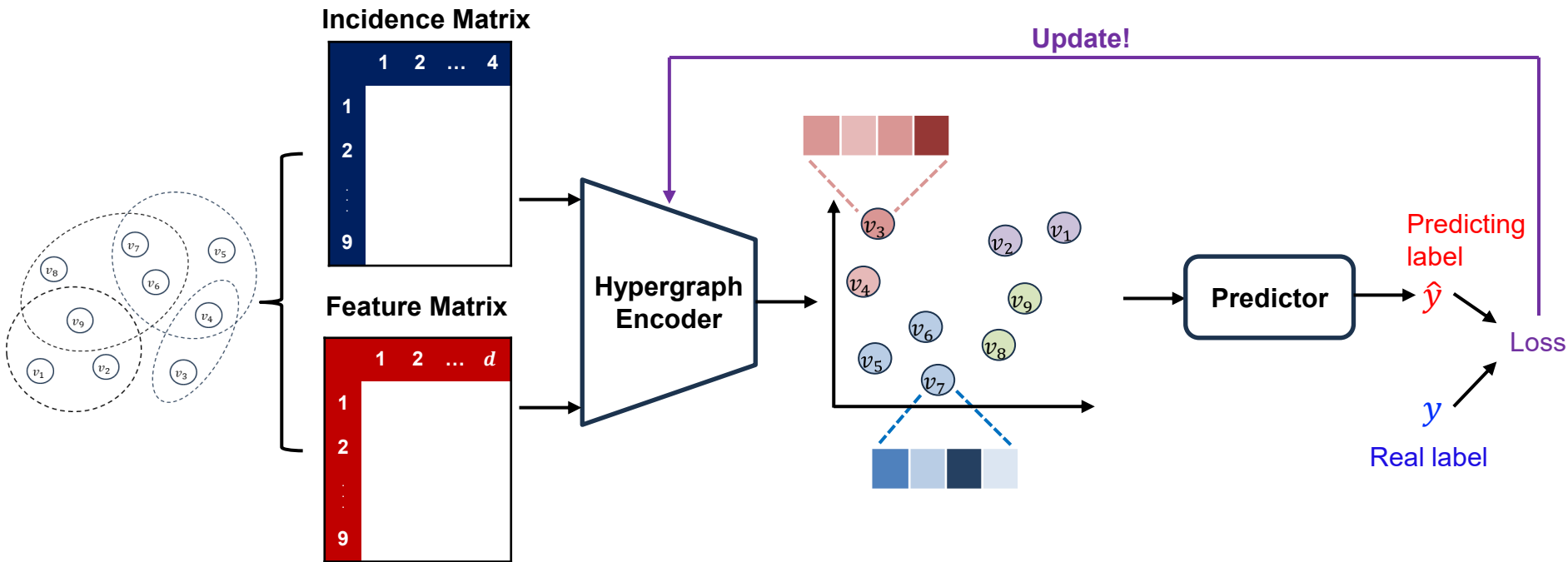
□ Hypergraphs can naturally model higher-order interactions as **hyperedges**

■ A **hyperedge** is subset of vertices



Hypergraph Neural Networks

- Learn embeddings by aggregating and transforming signals induced by incident hyperedges



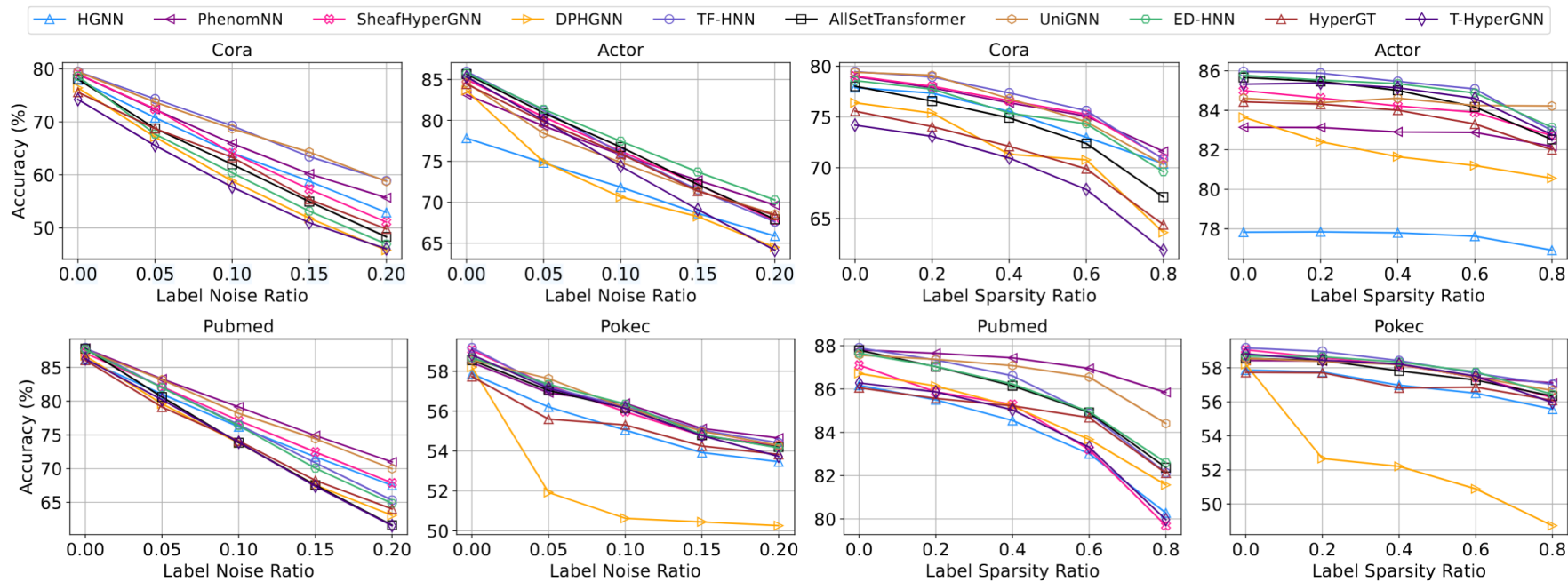
Problems

- Hypergraph data labeling is often **time, resource, and labor-intensive**



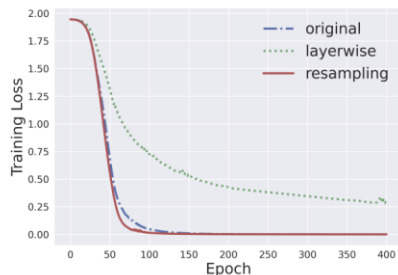
Label Dependence Bottleneck

Noisy or sparse labels degrade supervised HyperGNNs

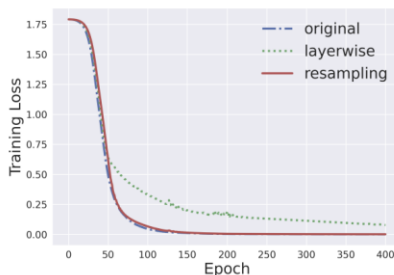


□ Label Dependence Bottleneck

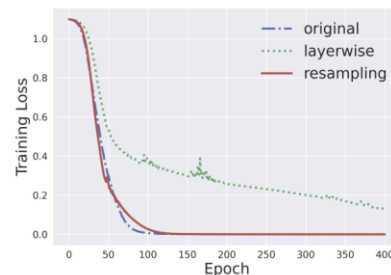
■ HNNs trained only in a supervised way can easily overfit



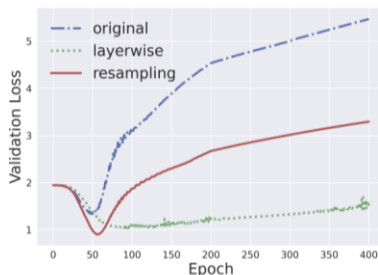
(a) Cora - Training



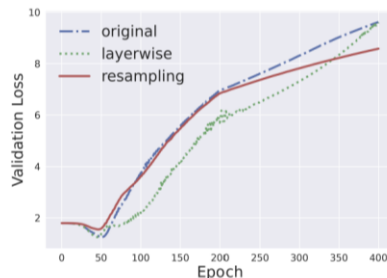
(b) CiteSeer - Training



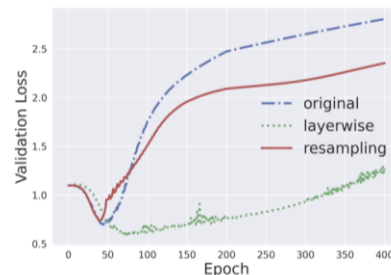
(c) PubMed - Training



(d) Cora - Validation



(e) CiteSeer - Validation

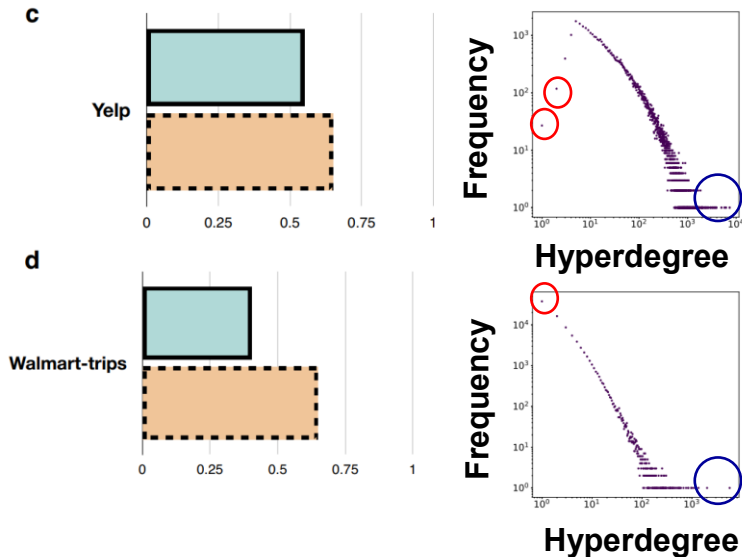
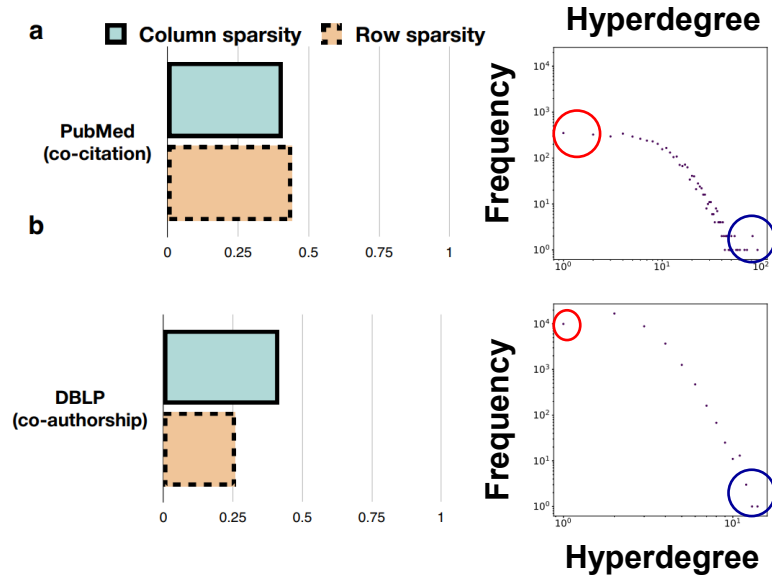


(f) PubMed - Validation

□ Structural Sparsity Bottleneck

■ Most nodes participate in only a few hyperedges

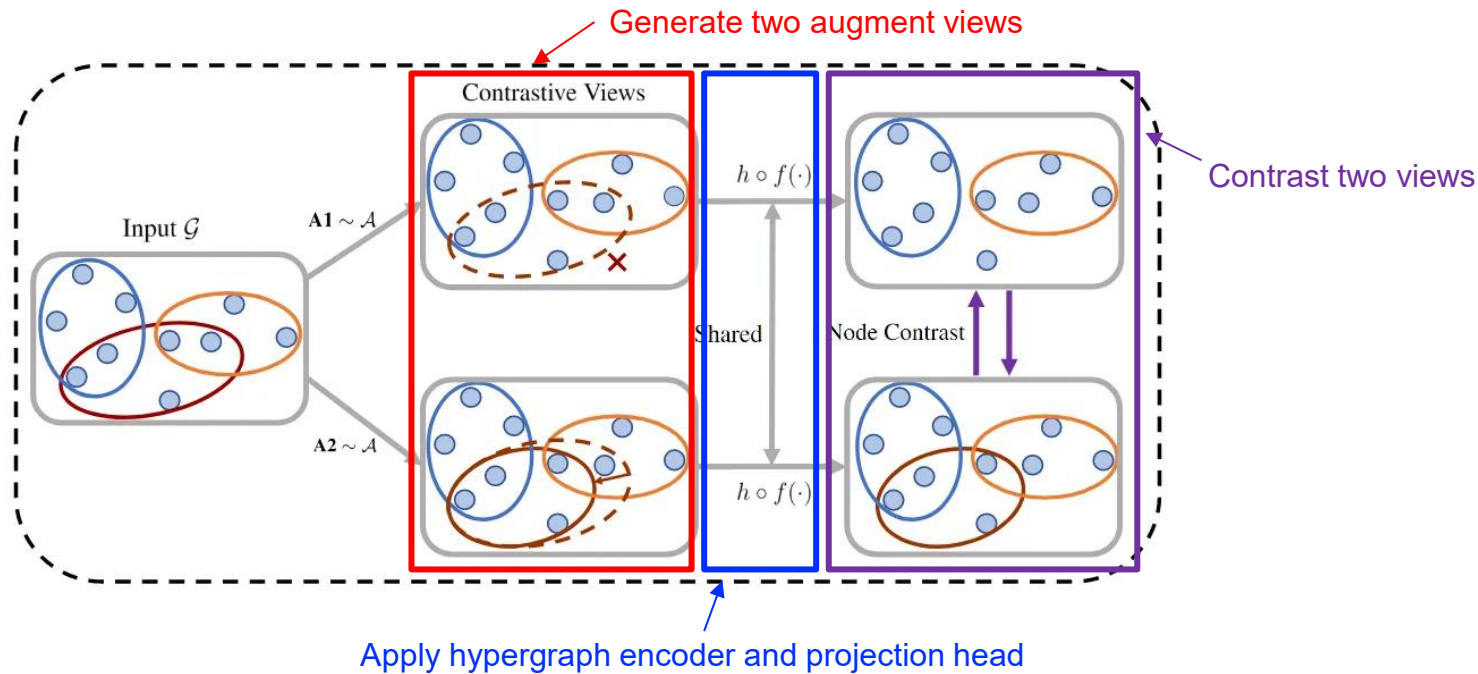
○ : Low-degree nodes
○ : High-degree nodes



Solution: Hypergraph Contrastive Learning (HGCL)

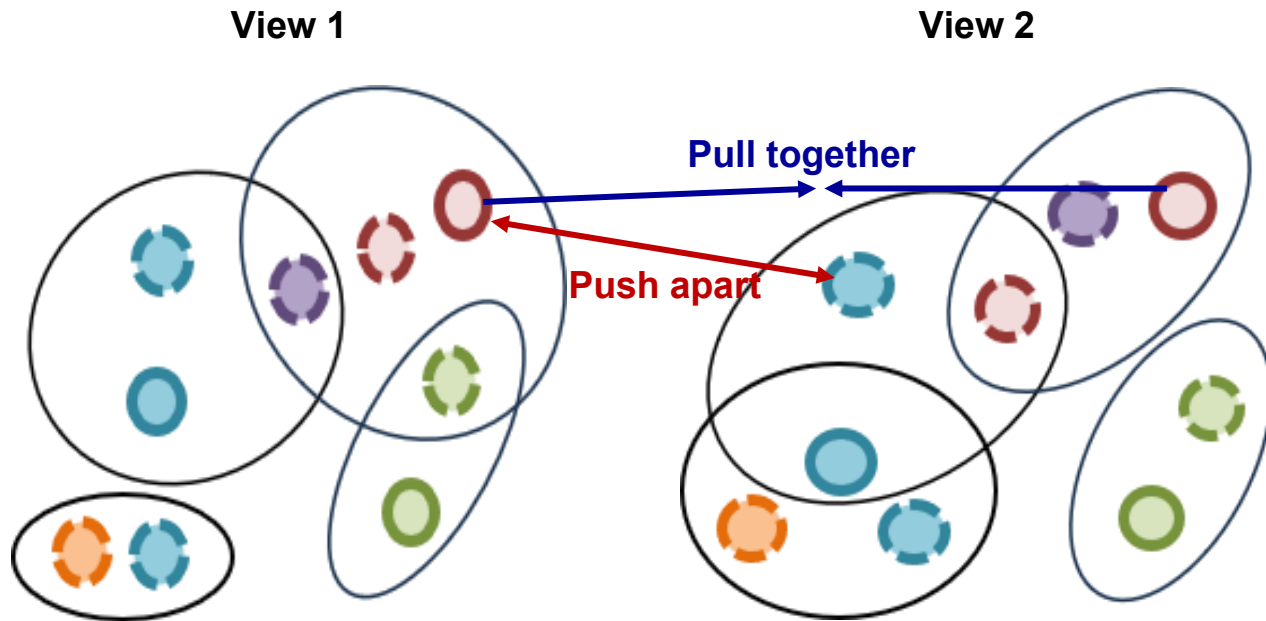
□ Contrast Augmented Views, Not Labels

- HGCL turns unlabeled structure into training signal by contrasting two augmented views



How HGCL Alleviates Label Sparsity

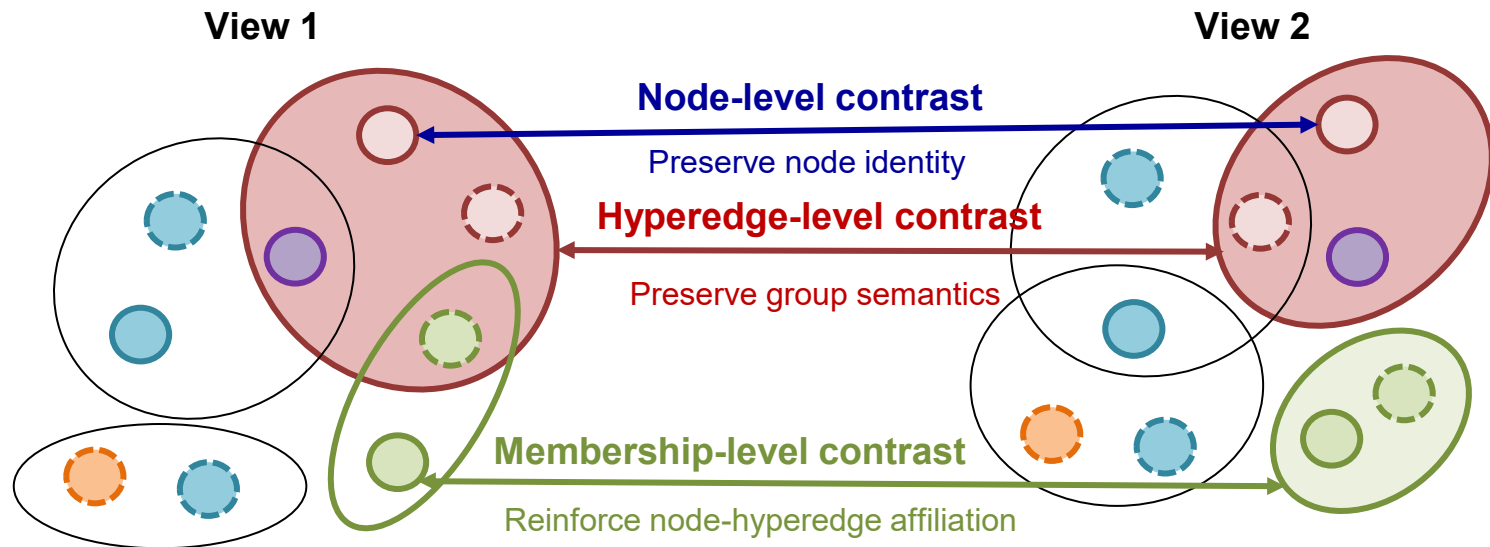
- Pull positive pair closer, **push negative pair away**



How HGCL Alleviates Structural Sparsity

- Multi-level contrast extracts richer signals from hypergraph structures

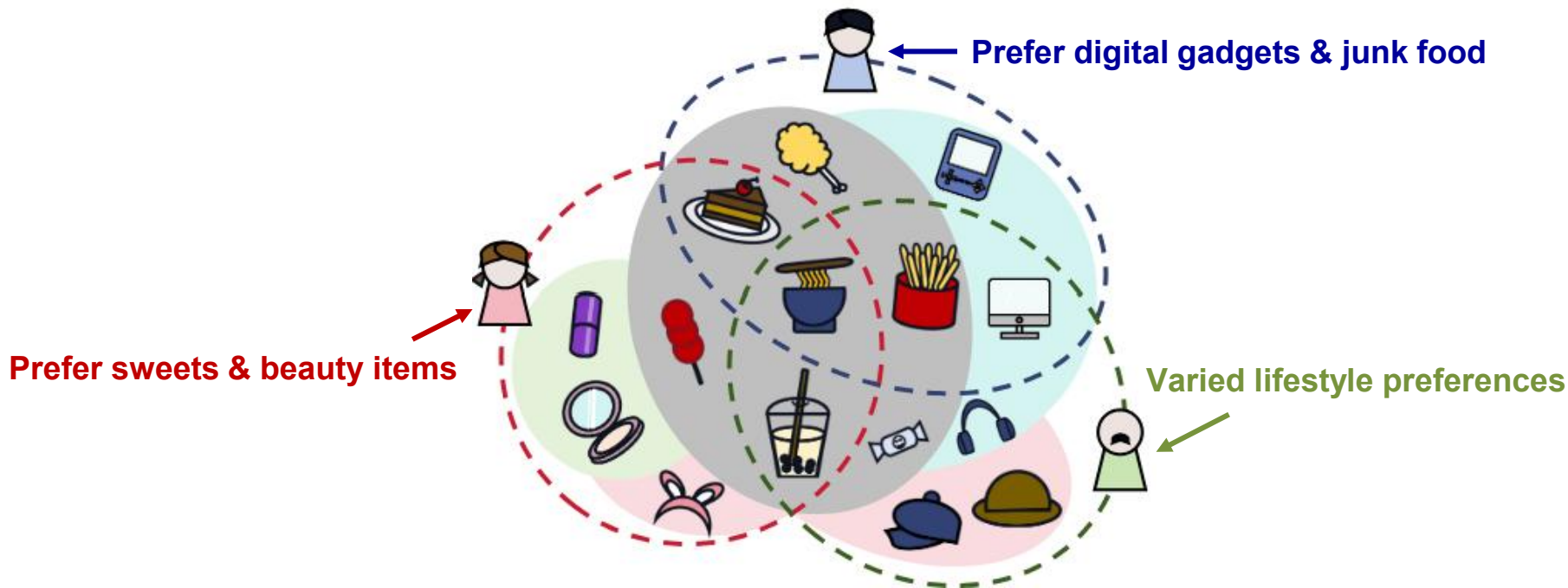
■ **Node:** identity · **Hyperedge:** group semantics · **Membership:** affiliation



Challenges

□ How to augment a hypergraph?

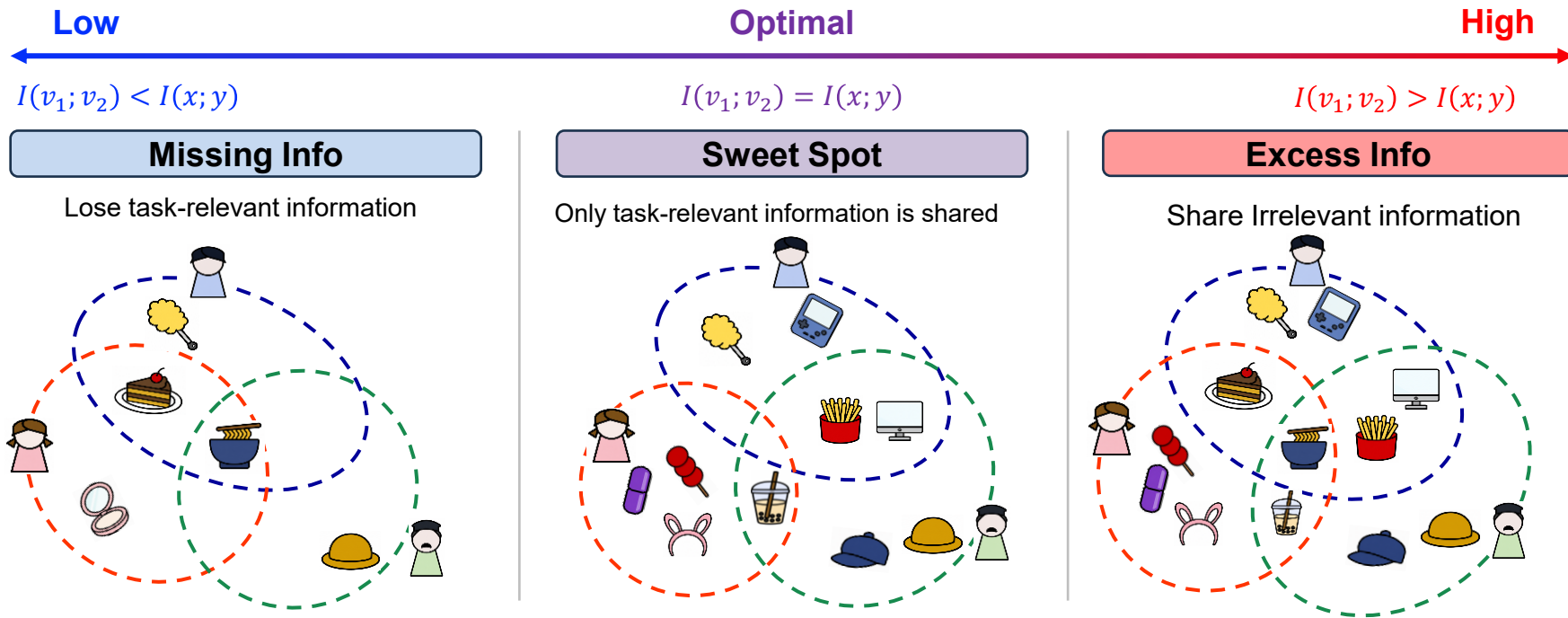
- The choice of views is what controls the information the representation captures



Challenges

□ How to augment a hypergraph?

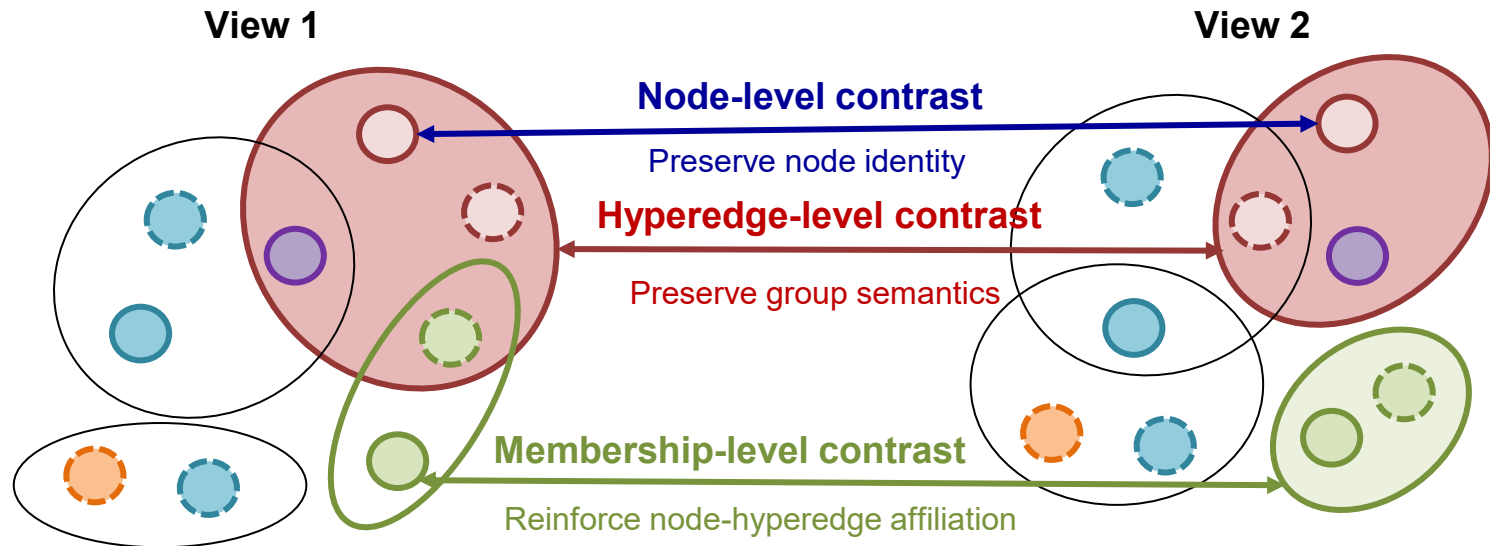
- The choice of views is what controls the information the representation captures



Challenges

□ What to contrast?

- Contrast targets determine the learned embedding structure



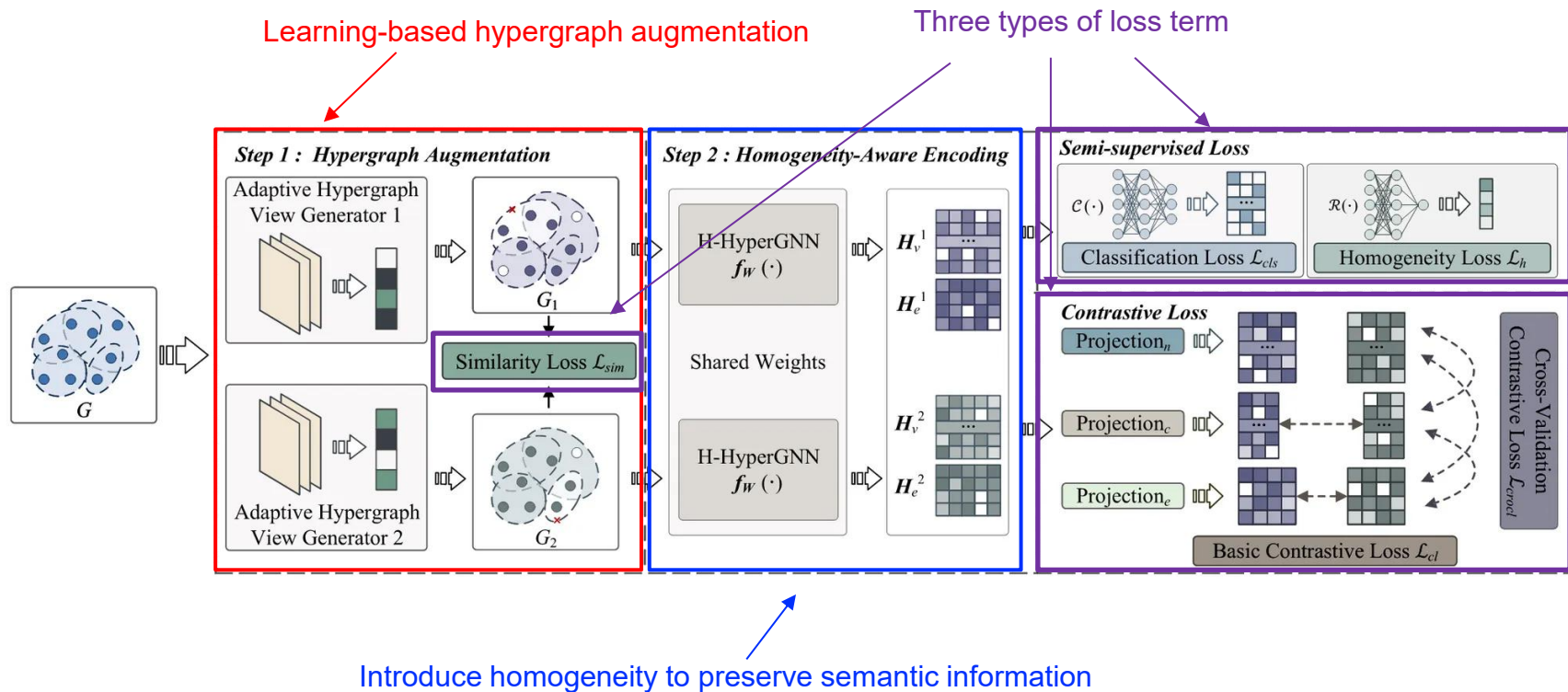


CHGNN: A Semi-Supervised Contrastive Hypergraph Learning Network

Yumeng Song, Yu Gu, Tianyi Li, Jianzhong Qi, Zhenghao Liu,
Christian S. Jensen, and Ge Yu

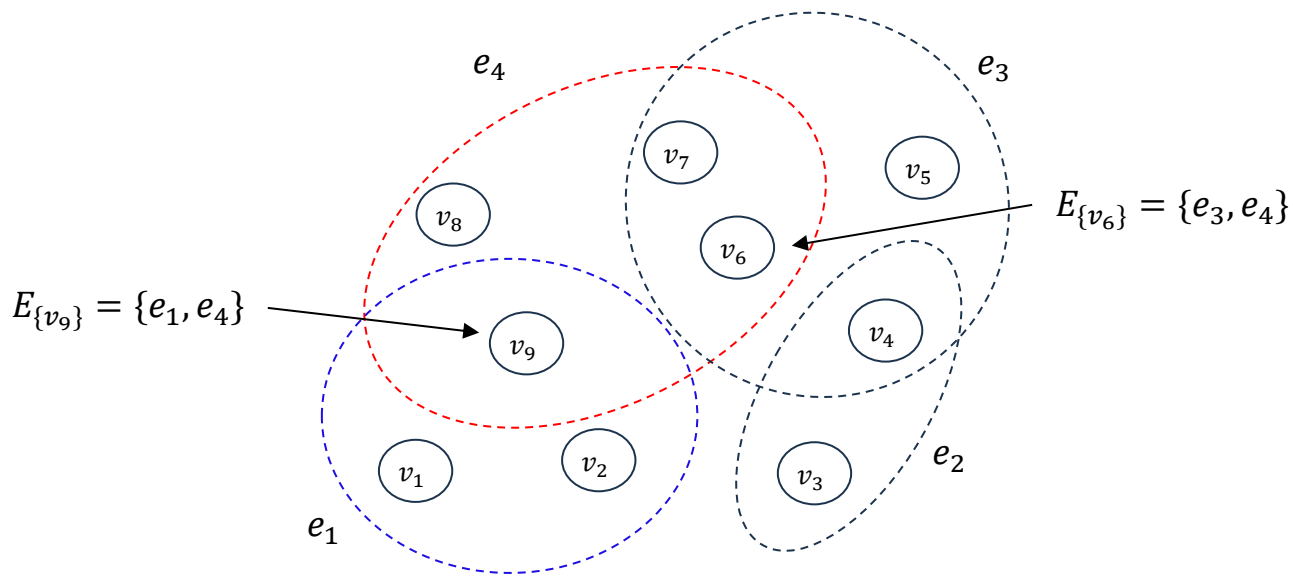
24-TKDE

Overview



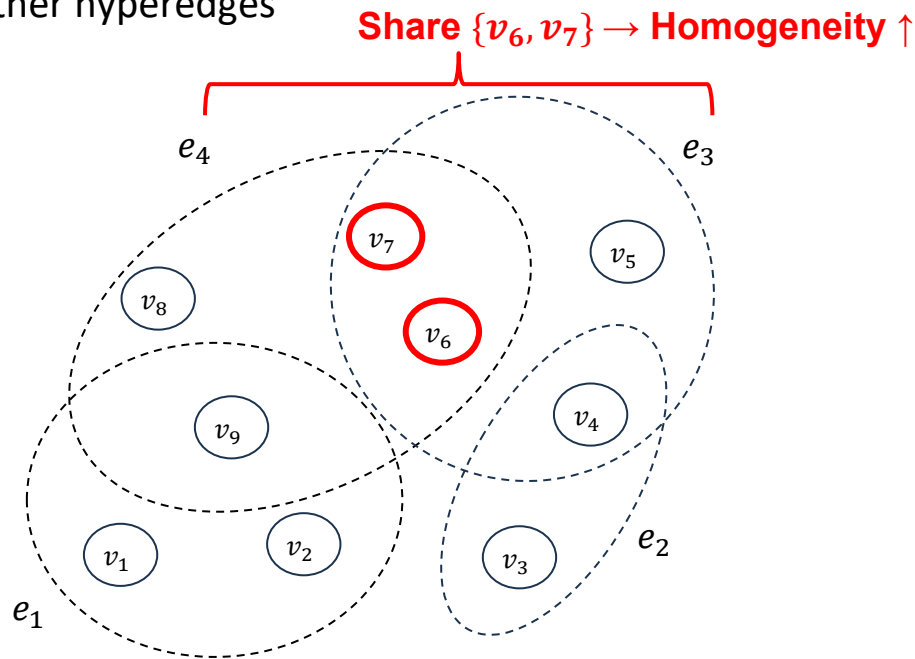
□ Egonet

- Given a hypergraph $G = (V, E)$, the **egonet** $E_{\{v_i\}}$ of a node $v_i \in V$ is the set of hyperedges that contain v_i



□ Homogeneity

- The larger the hyperedge homogeneity, the more often the node pairs within the hyperedge appear together in other hyperedges

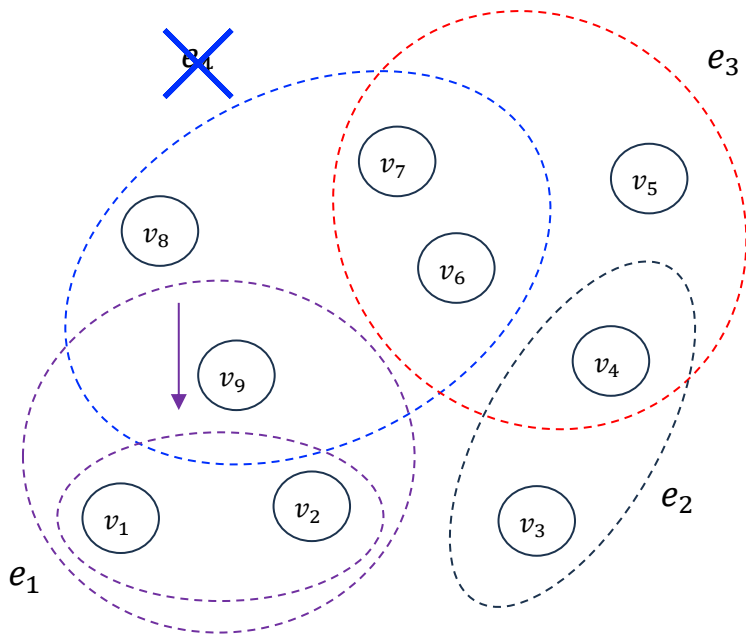


Hypergraph Augmentation

□ Adaptive Hypergraph View Generator

■ Augmentation operations

- Preserve hyperedges (P), Remove hyperedges (R), Mask some nodes in the hyperedges (M)

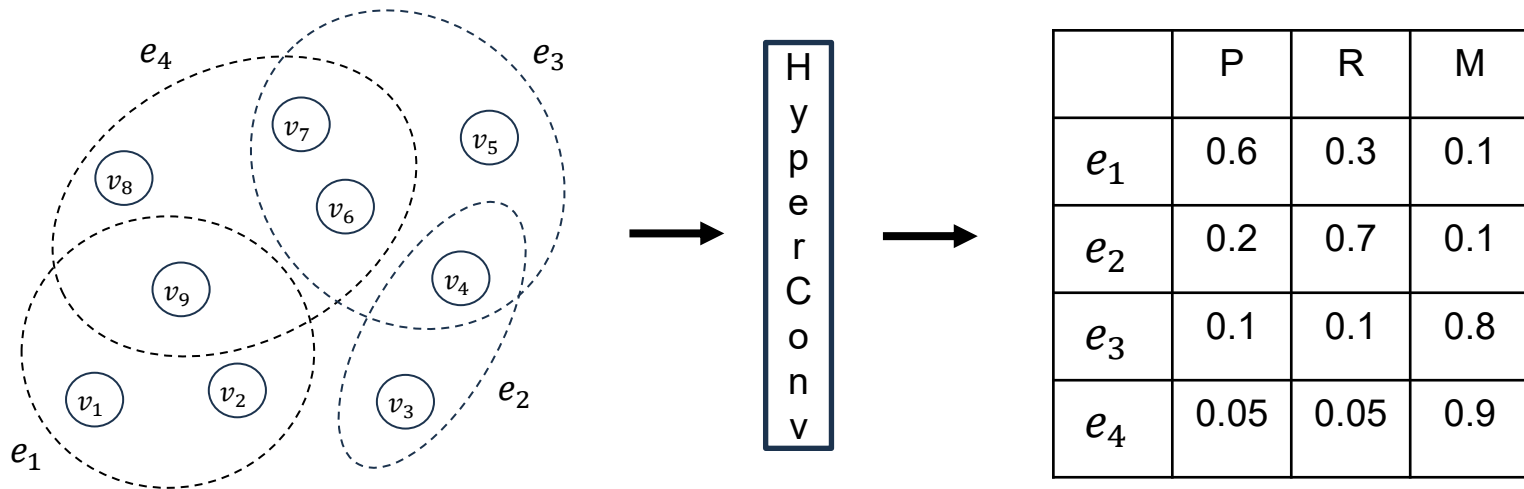


Hypergraph Augmentation

□ Adaptive Hypergraph View Generator

■ Obtain the hyperedge embedding matrix M_E using the HyperConv layers

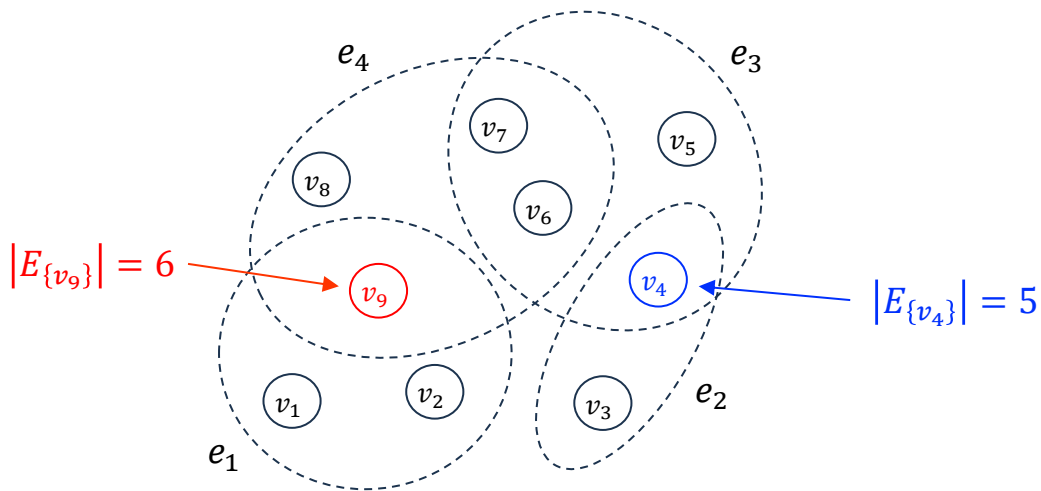
□ M_E : the probabilities of preserving, removing, and masking operations



Hypergraph Augmentation

□ Overlappness-Based Masking

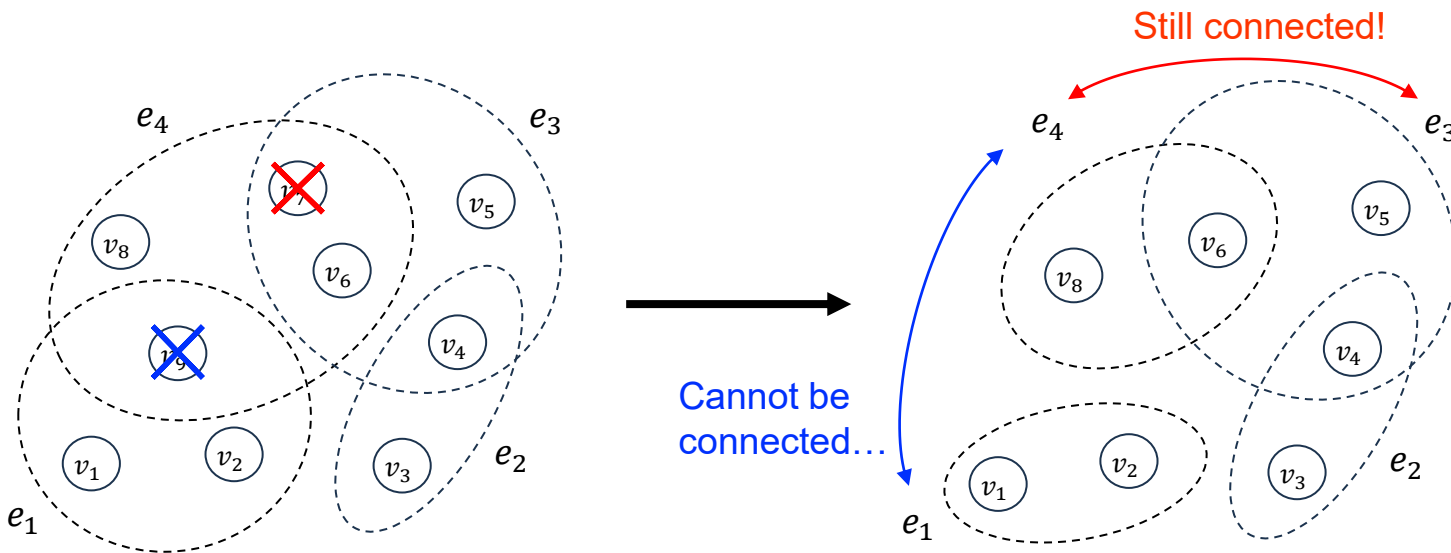
- A node is considered more important if its **egonet** contains more distinct nodes



Hypergraph Augmentation

□ Overlappness-Based Masking

- If the hyperedges contained in the egonet of a node overlap less with the hyperedges contained in the egonets of other nodes, the node is considered more important



Hypergraph Augmentation

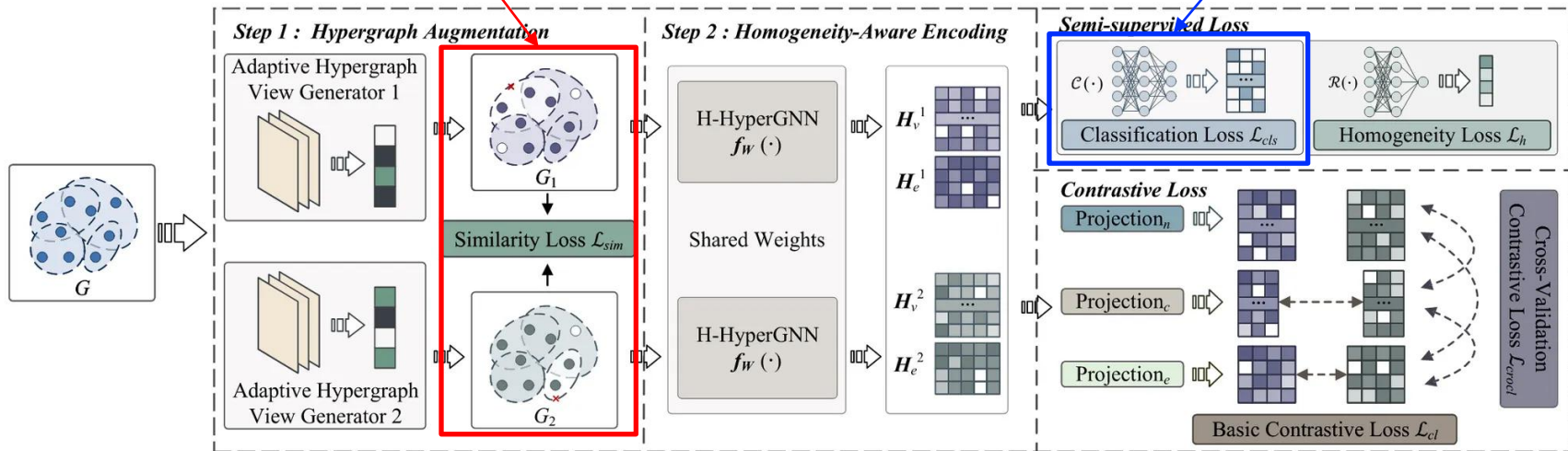
View Generator Training

■ Train a view generator in an end-to-end manner to generate task-dependent views

- Classification loss : Guide the generator to retain classification information
- Similarity loss : Minimize the mutual information between the views

Make two augmented views dissimilar

Make augmented view retain task-relevant information

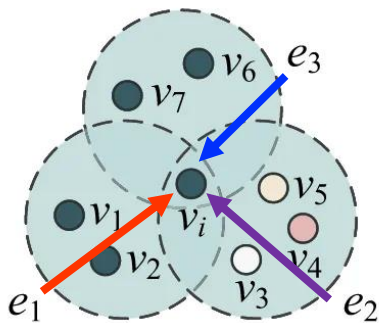


Hyperedge Homogeneity-Aware HyperGNN

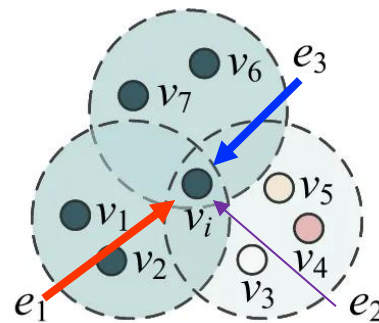
- Make the nodes of homogeneous hyperedge to be similar, the nodes of heterogeneous hyperedge to be dissimilar

- $$h_{e,e_i}^{(l)} = \frac{1}{|e|} \sum_{v_j \in e_i} h_{v,v_j}^{(l-1)}$$

- $$h_{v,v_i}^{(l)} = W \left(h_{v,v_i}^{(l-1)} + \sum_{e_j \in E_{\{v_i\}}} \text{homo}(e_j) \cdot h_{e,e_j}^{(l)} \right)$$



(a) Equal aggregation



(b) Ideal aggregation

□ Semi-Supervised Training Loss

■ Classification loss

- Make a view generator and hypergraph encoder learn classification information

- $\mathcal{L}_{cls} = -\sum_{v_i \in V_L} y_i \ln \mathcal{C}\left(\frac{h_{v,v_i}^1 + h_{v,v_i}^2}{2}\right)$

■ Hyperedge homogeneity loss

- Aim to **decrease the similarity between node embeddings in heterogeneous hyperedges** and **increase the similarity in homogeneous hyperedges**

- $\mathcal{L}_h = \lambda_h \cdot \left(\frac{1}{2} \sum_{i \in \{1,2\}} \text{MSE}(\mathcal{R}(H_e^i), v_{homo})\right)$

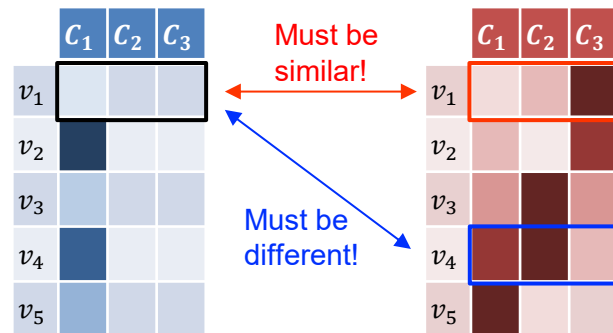
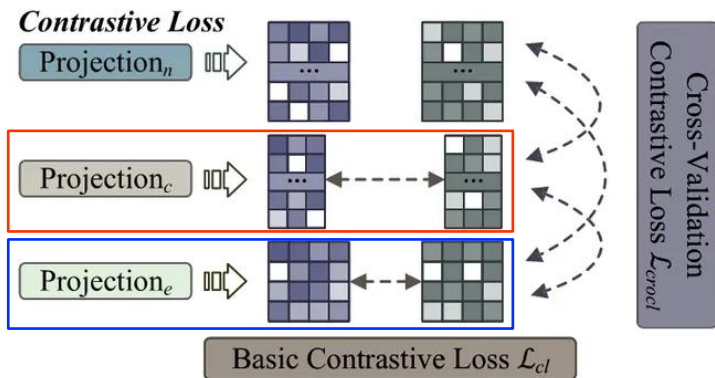
□ Basic Contrastive Loss

■ Cluster-level contrastive loss

- Focus on clustering nodes with similar categories and separating nodes with different categories

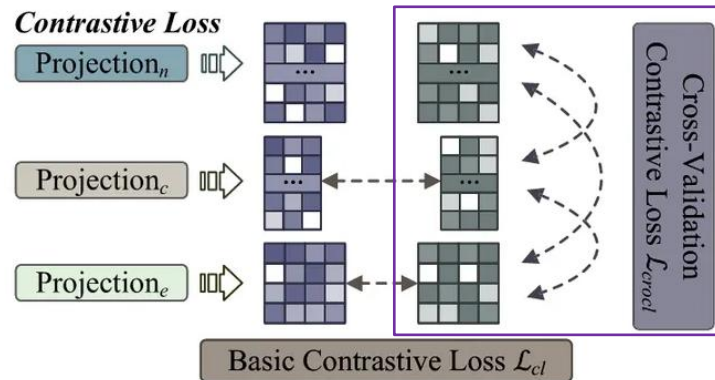
■ Hyperedge-level contrastive loss

- Aim to maximize the mutual information of hyperedge representations between different views



□ Cross-Validation Contrastive Loss

- Enable the embeddings of different types of data to obtain additional information beyond that contained in the data of the same type



□ Dataset

Dataset	#nodes	#hyperedges	#attributes	#classes	label ratio
CA-Cora	2,708	1,072	1,433	7	5.2%
CA-DBLP	43,413	22,535	1425	6	4.0%
CC-Pubmed	19,717	7,963	500	3	0.8%
CC-Citeseer	3,312	1,079	3,703	6	4.2%
CC-Cora	2,708	1,579	1,433	7	5.2%
20news	16,242	100	100	4	50%
ModelNet	12,311	12,311	100	40	50%
NTU	2,012	2,012	100	67	50%
Mushroom	8,124	298	22	2	50%

Comparison

Consistently outperform all competitors

	Dataset	CA-Cora	CA-DBLP	CC-Pubmed	CC-Citeseer	CC-Cora	20news	ModelNet	NTU	Mushroom
Non-Graph	MLP	57.8 ± 4	80.6 ± 0.8	73.3 ± 1.6	59.1 ± 4	59.9 ± 2.6	81.1 ± 1.1	95.9 ± 0.8	87.2 ± 2.8	98.4 ± 1.7
	SimCLR*	44.2 ± 1.6	76.9 ± 0.4	72.5 ± 1.4	45.5 ± 4.4	39.5 ± 1.3	80.7 ± 0.7	94.4 ± 3.9	82.8 ± 2.5	96.3 ± 0.3
	MoCo*	42.2 ± 2	75.9 ± 0.5	66.3 ± 1.5	43.8 ± 1.4	38.4 ± 1.5	80 ± 0.4	96.5 ± 2.2	82.1 ± 2	88.6 ± 0.8
	SimCLRv2	55 ± 5.3	80.1 ± 0.4	74.5 ± 2.8	57.8 ± 4	57.6 ± 3.7	81.4 ± 0.4	95.3 ± 0.2	85.9 ± 1.3	98.3 ± 0.2
	Suave	61.3 ± 2.8	80.8 ± 2.9	75.2 ± 2.3	62.8 ± 2.3	60.8 ± 3	81.5 ± 0.6	96.6 ± 0.3	89.1 ± 0.7	98.6 ± 0.8
Graph	GCN	63.7 ± 1.8	85.6 ± 0.3	74.6 ± 1	59.4 ± 2.3	60.4 ± 1.3	83.2 ± 0.4	95.9 ± 0.2	85.1 ± 0.9	93.6 ± 0.2
	GAT	65.7 ± 1.9	85.2 ± 0.3	74.8 ± 1.8	59.3 ± 1.7	60.2 ± 1.7	79.7 ± 7	95.5 ± 0.2	82.8 ± 1.4	90.3 ± 13
	DGI*	64.4 ± 0.3	79.3 ± 0.3	70.9 ± 0.2	59.7 ± 0.3	61.4 ± 0.3	82.8 ± 0.3	95.9 ± 0.2	79.5 ± 0.2	85.7 ± 0.1
	SimGRACE*	63.1 ± 1.5	82.2 ± 0.3	74.5 ± 1.2	57.2 ± 1.8	64.3 ± 2	82.5 ± 0.5	95.8 ± 0.3	76.6 ± 1.4	90 ± 0.3
	HomoGCL*	58.3 ± 1.6	OOM	74.8 ± 1	55.8 ± 1.2	54.3 ± 1.5	82.4 ± 0.3	95.7 ± 0.2	74.3 ± 0.7	91.7 ± 0.3
	MA-GCL*	68.5 ± 1.9	85.6 ± 0.2	74.1 ± 1.9	65.8 ± 1.3	68.9 ± 1.7	82.9 ± 0.4	93.8 ± 0.3	80.3 ± 1.1	92.5 ± 0.3
Hypergraph	HyperConv	65.2 ± 1.5	88.3 ± 0.1	40.1 ± 7	37.5 ± 3.3	43.9 ± 3.3	81 ± 0.5	94.3 ± 0.1	86.1 ± 0.8	97.9 ± 0.2
	HGNN	69.9 ± 3.3	78.5 ± 0.3	75.3 ± 1.1	60.7 ± 2.2	62.7 ± 3.7	81.1 ± 0.6	95.3 ± 0.6	86.2 ± 2.3	96.7 ± 0.9
	HyperGCN	72.1 ± 5	88.3 ± 0.6	73.1 ± 9.6	60.1 ± 3.6	62.5 ± 3.5	81.3 ± 0.5	73.7 ± 13.2	53 ± 8.7	46.4 ± 0.1
	HNHN	65.5 ± 4.5	87.6 ± 0.7	70.7 ± 2.3	55.3 ± 2.4	62.4 ± 3	80.9 ± 0.5	97.5 ± 0.4	86.5 ± 5.1	98 ± 1.2
	HyperSAGE	72.9 ± 3.1	87.6 ± 0.8	73.1 ± 1.4	61.3 ± 2.6	67.4 ± 3.8	81.8 ± 0.6	97.5 ± 0.8	85.4 ± 3.8	97.8 ± 1.9
	UniGNN	74.3 ± 1.9	89.0 ± 0.3	76 ± 1.4	62.6 ± 3.2	68.3 ± 2.6	83.3 ± 1.1	97 ± 0.3	85.8 ± 2.8	98.1 ± 1.1
	AST	71.1 ± 3.8	87.7 ± 0.5	64.2 ± 9.8	60.3 ± 4.8	64.5 ± 3.3	82.5 ± 0.9	97.9 ± 0.5	88.1 ± 4.6	98.9 ± 1.7
	TriCL*	77.6 ± 1.1	OOM	76.7 ± 1.3	66.2 ± 1.6	73.2 ± 0.8	80.8 ± 0.5	97.6 ± 0.1	90.5 ± 0.7	99.7 ± 0.1
	CHGNN	78.9 ± 0.7	89.6 ± 1.5	79.8 ± 1.0	69.5 ± 1.1	75.3 ± 0.8	83.7 ± 0.2	98.1 ± 0.1	91.9 ± 0.2	99.8 ± 0.1

□ Effect of Label Ratios

■ Show high data efficiency

Dataset		CC-Cora									NTU							
	Ratio	0.5%	1%	2%	3%	10%	20%	30%	40%	50%	3%	4%	5%	10%	20%	30%	40%	80%
Non-Graph	MLP	31.1	39.9	45.7	52.6	63.8	68.5	70.5	72.7	73.2	42.0	50.2	55.3	65.1	77.9	79.9	84.9	91.1
	SimCLR*	30.1	39.6	44.7	47.1	54.0	56.0	56.8	57.2	59.9	31.5	31.3	35.7	56.9	58.3	59.6	77.2	82.9
	MoCo*	24.3	37.3	41.0	43.5	50.5	52.6	53.3	54.5	55.5	31.2	31.1	32.1	51.4	51.6	61.3	77.6	84.2
	SimCLRv2	30.3	39.1	45.0	48.9	64.0	68.5	70.6	72.8	73.7	36.7	50.8	55.6	69.6	78.6	80.2	81.3	87.1
	Suave	37.7	42.1	53.9	58.3	64.4	65.9	68.9	67.5	68.7	50.4	54.9	61.0	73.5	80.4	83.8	86.8	92.4
Graph	GCN	21.3	36.7	47.2	53.5	66.4	69.7	72.0	73.2	74.3	46.0	50.6	53.0	60.7	64.4	65.6	67.0	90.4
	GAT	24.8	39.4	49.0	53.4	66.0	70.9	72.1	73.0	74.3	45.8	53.4	53.3	59.4	63.8	64.9	66.4	88.5
	DGI*	26.8	30.2	38.8	43.3	62.0	63.8	63.6	63.2	62.3	47.6	56.3	62.2	65.4	73.3	74.5	79.8	91.2
	SimGRACE*	24.9	31.4	37.1	41.4	62.9	62.1	61.8	62.6	62.7	48.9	56.6	61.5	65.4	73.6	77.7	78.1	87.2
	HomoGCL*	32.2	37.7	42.0	45.1	51.4	53.0	54.4	55.5	56.5	50.8	50.8	57.2	58.8	64.9	67.6	70.9	84.1
	MA-GCL*	45.5	51.0	54.9	57.4	62.1	63.5	64.6	65.6	66.5	<u>56.9</u>	56.9	64.3	69.0	74.0	76.3	78.1	85.6
Hypergraph	HyperConv	27.2	30.8	36.6	43.2	50.0	55.9	54.2	62.5	66.1	55.6	57.7	<u>64.7</u>	67.4	81.3	83.1	84.6	90.1
	HGNN	30.2	40.8	56.7	60.4	71.6	75.8	77.1	77.3	78.6	28.3	31.4	29.5	38.6	43.6	45.6	48.7	89.2
	HNHN	31.7	44.9	52.6	56.2	68.0	71.1	72.6	74.6	74.9	43.2	47.4	51.1	65.4	74.7	78.0	79.5	88.5
	HyperSAGE	39.9	47.0	54.9	58.4	68.9	73.4	76.4	77.0	77.7	41.5	47.1	52.8	62.8	72.5	77.8	78.2	87.0
	UniGNN	38.5	49.0	58.5	62.5	71.7	74.5	76.8	77.1	78.4	47.8	55.5	58.7	71.6	79.1	82.2	84.4	86.5
	AST	48.6	49.8	55.5	58.9	69.5	71.4	75.2	77.9	78.4	40.5	44.5	52.5	66.0	75.3	75.4	78.9	83.9
	TriCL*	<u>53.3</u>	<u>66.6</u>	<u>69.8</u>	<u>73.5</u>	76.9	<u>78.6</u>	<u>78.2</u>	<u>78.5</u>	<u>79.0</u>	53.6	<u>60.8</u>	63.9	<u>76.4</u>	<u>83.4</u>	<u>86.5</u>	<u>87.9</u>	<u>92.5</u>
	CHGNN	64.8	66.7	72.2	73.9	76.9	78.7	78.5	79.7	79.9	57.7	63.8	68.8	77.8	84.8	86.8	88.3	94.1

□ Ablation Study

	$\mathcal{L}_h$	$\mathcal{L}_{cl}$	$\mathcal{L}_{crocl}$	Aug	Encoder	CA-Cora	CA-DBLP	CC-Pubmed	CC-Citeseer	CC-Cora
V_A	-	-	-	-	HyperGNN	74.0 ± 0.4	87.9 ± 0.1	75.1 ± 0.8	62.0 ± 0.8	66.8 ± 1.5
V_B	✓	-	-	-	HyperGNN	73.1 ± 0.7	87.1 ± 0.1	74.9 ± 0.3	60.3 ± 0.7	64.8 ± 1.3
V_C	-	-	-	-	H-HyperGNN	74.5 ± 0.1	88.1 ± 0.1	75.4 ± 0.9	61.6 ± 1.3	67.3 ± 1.3
V_D	✓	-	-	-	H-HyperGNN	74.7 ± 0.1	88.2 ± 0.1	75.9 ± 1.1	62.5 ± 0.2	67.4 ± 1.2
V_E	-	✓	-	RandAug	HyperGNN	71.8 ± 1.6	88.5 ± 0.2	74.6 ± 1.6	59.1 ± 3.2	63.1 ± 2.6
V_F	-	-	✓	RandAug	HyperGNN	75.5 ± 0.8	88.7 ± 0.2	75.8 ± 3.6	68.2 ± 1.7	72.0 ± 1.6
V_G	-	✓	✓	RandAug	HyperGNN	74.0 ± 0.9	88.6 ± 0.2	75.4 ± 4.0	67.7 ± 1.1	70.5 ± 2.2
V_H	✓	✓	-	RandAug	H-HyperGNN	70.3 ± 2.3	88.6 ± 0.2	74.1 ± 1.7	58.9 ± 1.9	63.7 ± 1.3
V_I	✓	-	✓	RandAug	H-HyperGNN	75.3 ± 1.2	88.8 ± 0.2	76.7 ± 3.4	67.5 ± 1.5	71.5 ± 0.6
V_J	✓	✓	✓	RandAug	H-HyperGNN	74.5 ± 1.9	88.5 ± 0.3	76.1 ± 2.6	67.0 ± 2.3	69.9 ± 3.8
V_K	-	✓	-	ViewGen	HyperGNN	75.7 ± 1.0	89.2 ± 0.1	75.5 ± 1.5	62.6 ± 1.5	66.3 ± 2.6
V_L	-	-	✓	ViewGen	HyperGNN	76.5 ± 1.3	89.1 ± 0.2	77.9 ± 1.7	67.6 ± 1.8	73.1 ± 0.9
V_M	-	✓	✓	ViewGen	HyperGNN	76.8 ± 1.4	89.1 ± 0.1	77.7 ± 1.8	67.9 ± 1.2	72.6 ± 1.3
V_N	✓	✓	-	ViewGen	H-HyperGNN	78.1 ± 1.2	89.1 ± 0.2	75.3 ± 1.3	62.1 ± 1.2	71.0 ± 2.4
V_O	✓	-	✓	ViewGen	H-HyperGNN	77.5 ± 1.0	89.1 ± 0.2	76.4 ± 1.0	68.0 ± 1.8	74.3 ± 0.7
V_P	✓	✓	✓	ViewGen	H-HyperGNN	78.3 ± 1.0	89.3 ± 0.2	78.3 ± 2.9	68.5 ± 1.7	75.5 ± 0.9

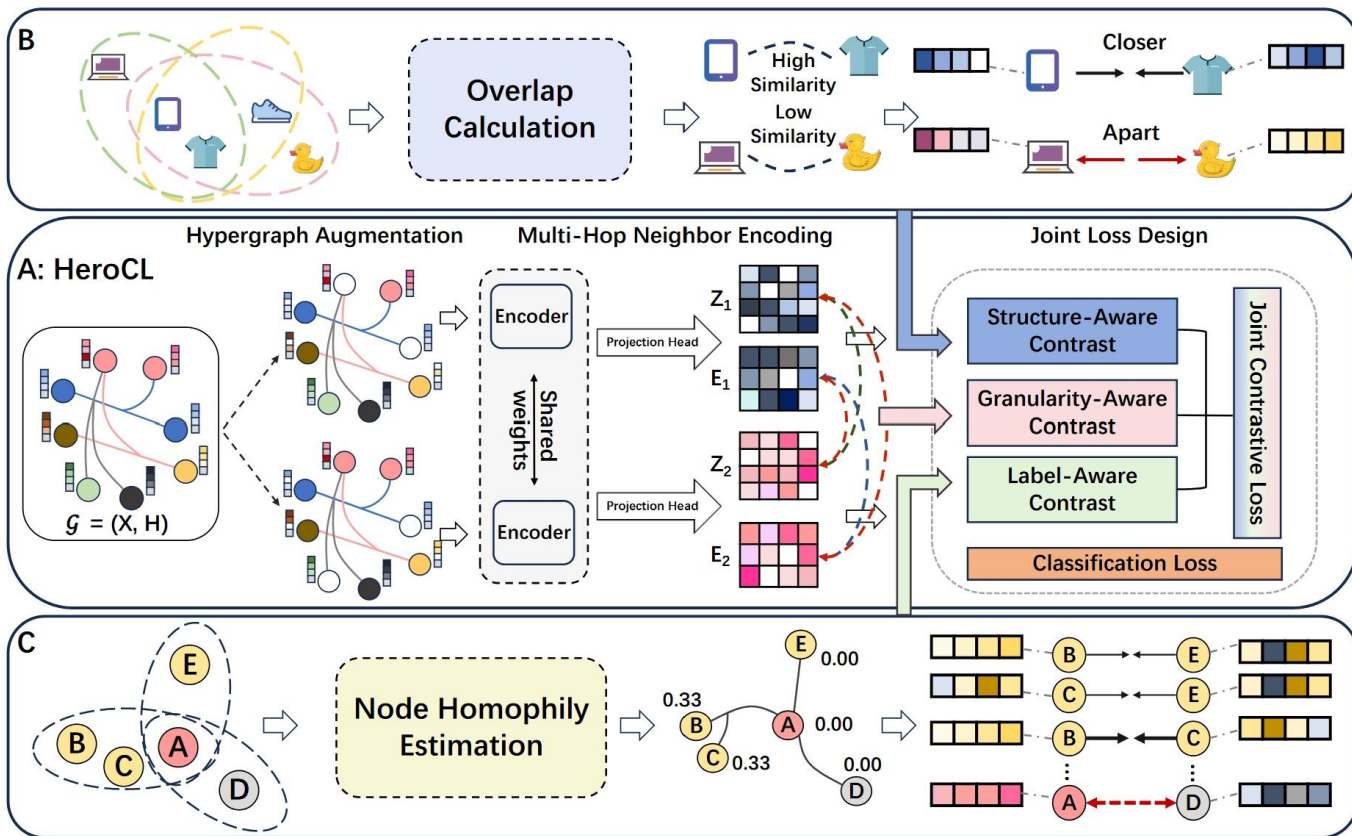


Heterophily-aware Contrastive Learning for Heterophilic Hypergraphs

Ming Li, Yongqi Li, Yuting Chen, Feilong Cao, Ke Lv

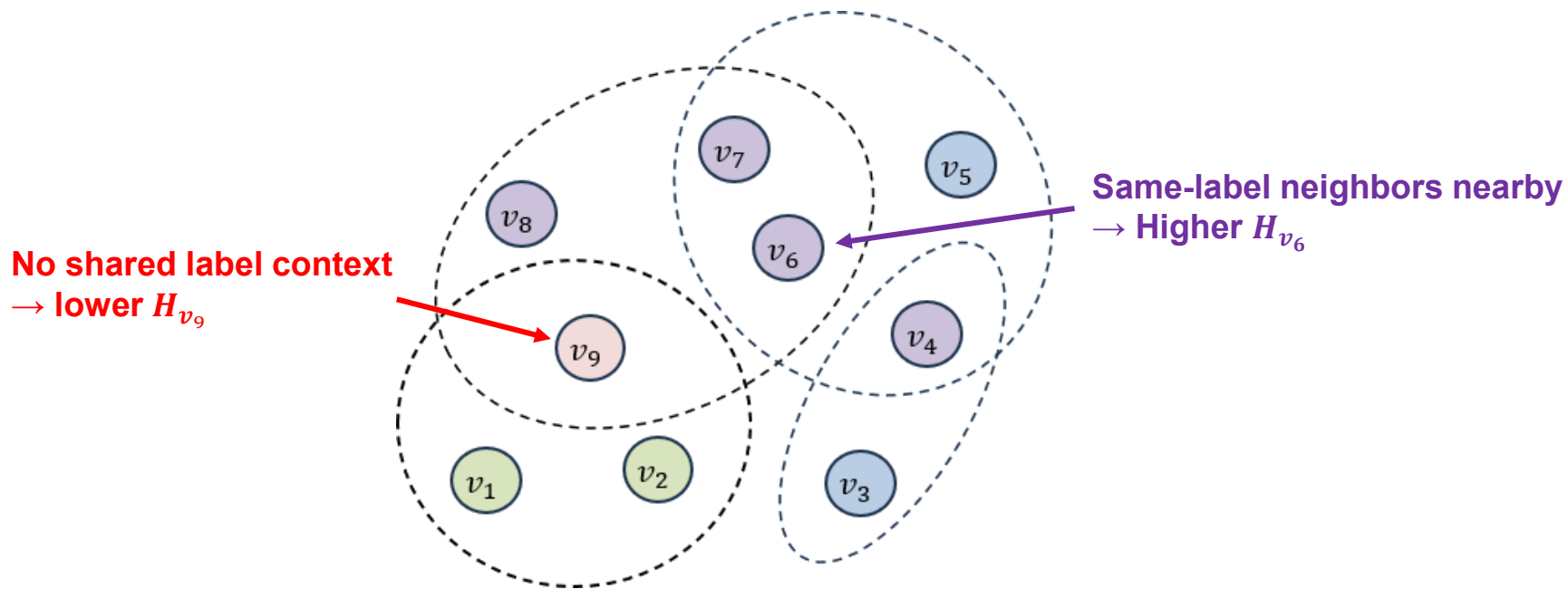
26-AAAI

Overview



□ Node Homophily

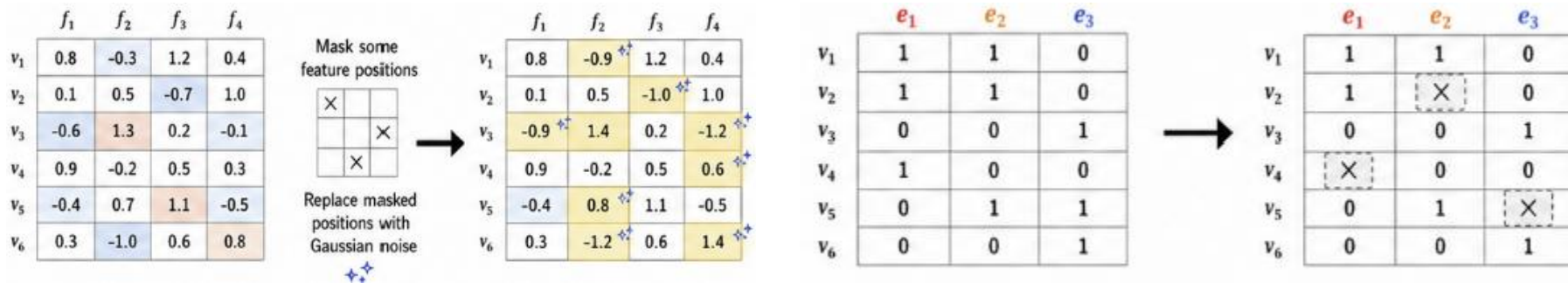
- Measure label consistency within a node's incident hyperedges



Hypergraph Augmentation

□ Generate two augmented views by applying perturbations to both **node features** and **hypergraph structure**

- Randomly mask positions in the node feature matrix and inject Gaussian noise
- Drop incidence



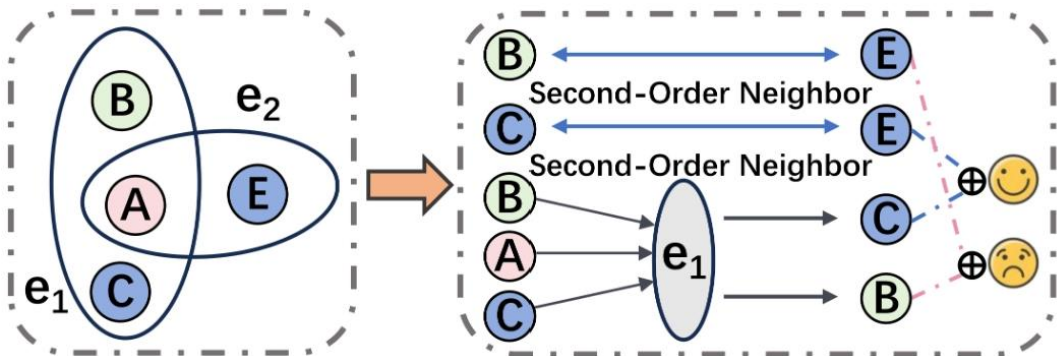
Multi-Hop Neighbor Encoding

□ Capture useful indirect context beyond noisy local neighbors

■ First-Order Propagation

$$\square \mathbf{X}_e^{(l)} = \mathbf{D}_e^{-1} \mathbf{H}^\top \mathbf{X}^{(l-1)}$$

$$\square \mathbf{X}_v^{(l)} = \mathbf{D}_v^{-1} \mathbf{H} \mathbf{X}_e^{(l)}$$



■ Second-Order Aggregation

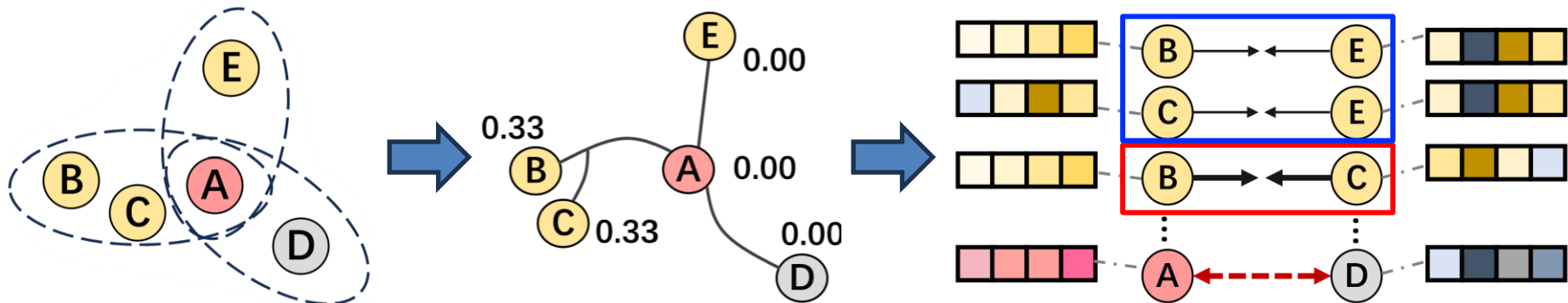
$$\square \mathbf{X}_v^{(l)} = \alpha_{2hop} \cdot \mathbf{X}_v^{(l)} + (1 - \alpha_{2hop}) \cdot \mathbf{X}_{2hop}^{(l)}$$

$$\square \mathbf{X}_{2hop}^{(l)} = \widehat{\mathbf{D}}_2^{-1} \left(\widehat{\mathbf{A}}^{(2)} \odot \mathbf{S} \right) \widehat{\mathbf{D}}_2^{-1} \mathbf{X}^{(l-1)} \mathbf{W}$$

Label-Aware Contrastive Learning

□ Encourage nodes sharing the same label to have similar embeddings

■ $\mathcal{L}_\ell = \frac{1}{2|V_l|} \sum_{i \in V_l} \sum_{j \in V_l} W_{i,j} \left(\ell_l^{(1)} \left(z_i^{(1)}, z_j^{(1)} \right) + \ell_l^{(2)} \left(z_i^{(2)}, z_j^{(2)} \right) \right)$

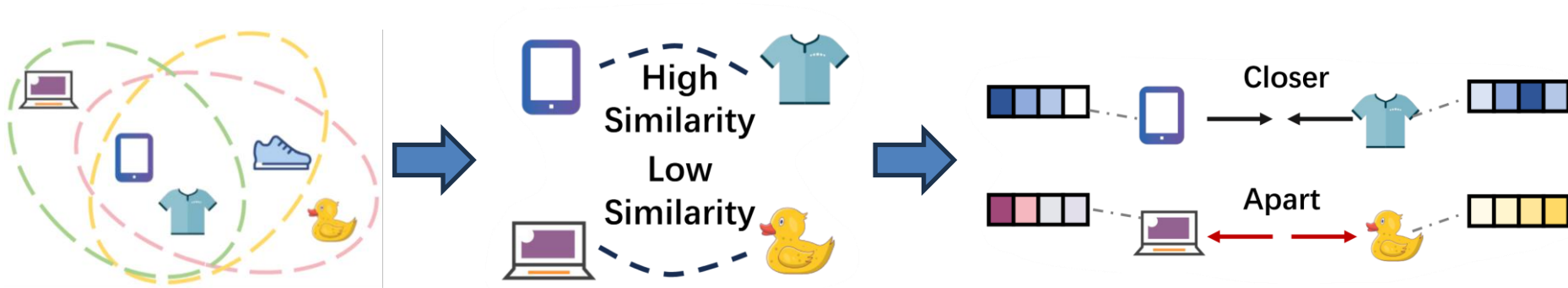


$H_i \backslash H_j$	High	low
High	Similar homophily $\rightarrow$ High $W_{i,j}$	Different homophily $\rightarrow$ Low $W_{i,j}$
low	Different homophily $\rightarrow$ Low $W_{i,j}$	Similar homophily $\rightarrow$ High $W_{i,j}$

Structure-Aware Contrastive Learning

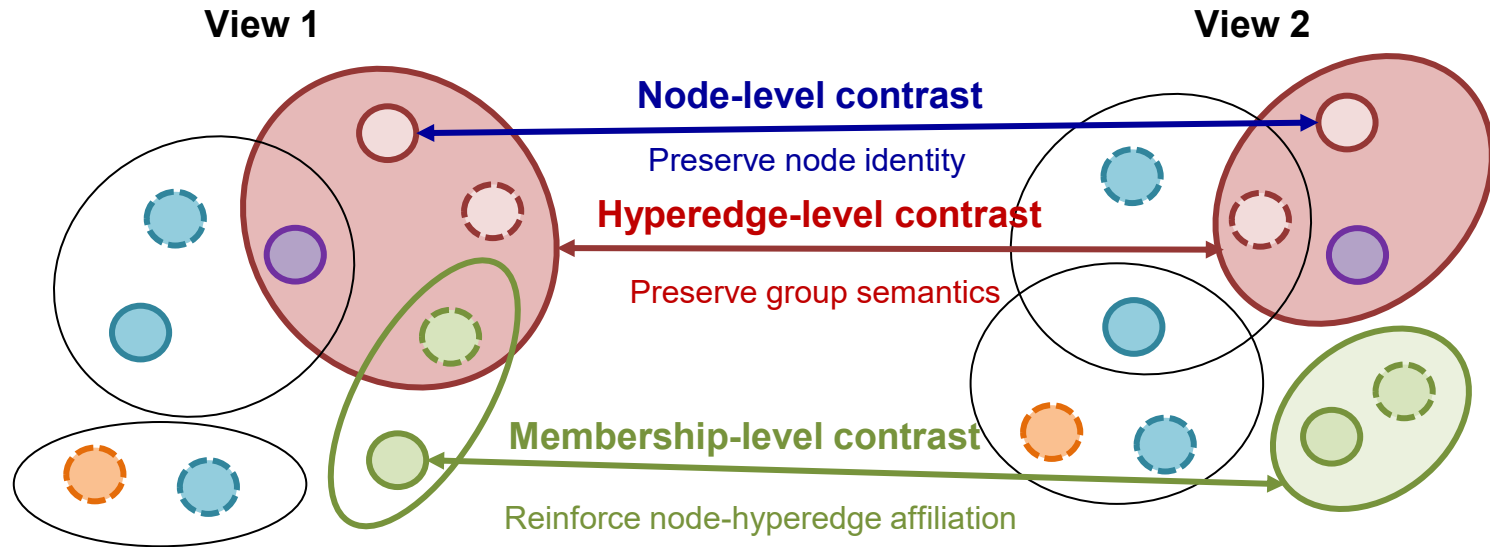
- Encourage structurally aligned nodes to have similar embeddings

- $\mathcal{L}_s = \frac{1}{2|P|} \sum_{(i,j) \in P} W_{i,j} \left(\ell_s^{(1)} \left(z_i^{(1)}, z_j^{(1)} \right) + \ell_s^{(2)} \left(z_i^{(2)}, z_j^{(2)} \right) \right)$



Granularity-Aware Contrastive Learning

- Enhance representation consistency across different levels of the hypergraph structure



□ Comparison on homophilic datasets

Model	Cora (0.75)	Citeseer (0.68)	Pubmed (0.78)	Cora-CA (0.78)	DBLP-CA (0.87)	Ave. Rank
MLP	75.16 ± 1.41	71.71 ± 1.01	87.20 ± 0.34	75.17 ± 1.41	84.37 ± 0.33	10.0
HGNN	79.39 ± 1.36	72.45 ± 1.16	86.44 ± 0.44	82.64 ± 1.65	91.03 ± 0.20	7.60
HyperGCN	78.45 ± 1.26	71.28 ± 0.82	82.84 ± 8.67	79.48 ± 2.08	89.38 ± 0.25	10.0
UniGCNII	78.81 ± 1.05	73.05 ± 2.21	88.25 ± 0.40	83.60 ± 1.14	91.69 ± 0.19	5.40
HyperND	79.20 ± 1.14	72.62 ± 1.49	86.68 ± 0.43	80.62 ± 1.32	90.35 ± 0.26	8.00
AllDeepSets	76.88 ± 1.80	70.83 ± 1.63	88.75 ± 0.33	81.97 ± 1.50	91.27 ± 0.27	7.80
AllSetTransformer	78.58 ± 1.47	73.08 ± 1.20	88.72 ± 0.37	83.63 ± 1.47	91.53 ± 0.23	5.80
ED-HNN	80.31 ± 1.35	73.70 ± 1.38	89.03 ± 0.53	83.97 ± 1.55	91.90 ± 0.19	3.00
SheafHyperGNN	81.30 ± 1.70	74.71 ± 1.23	87.68 ± 0.60	85.52 ± 1.28	91.59 ± 0.24	4.00
HyperUFG	81.51 ± 0.99	74.72 ± 2.10	88.73 ± 0.42	85.18 ± 0.69	91.67 ± 0.31	2.80
HeroCL (Ours)	83.03 ± 0.98	74.96 ± 0.35	90.99 ± 1.01	86.37 ± 1.08	91.61 ± 0.21	1.60

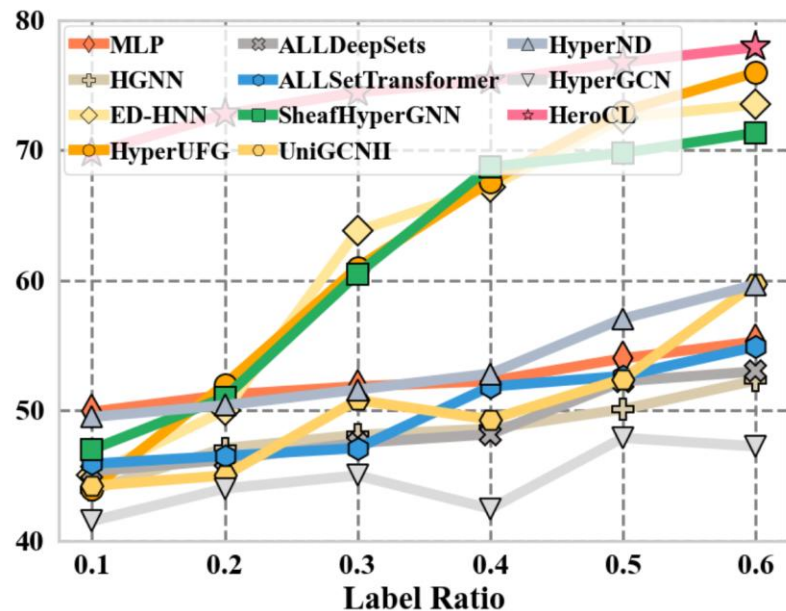
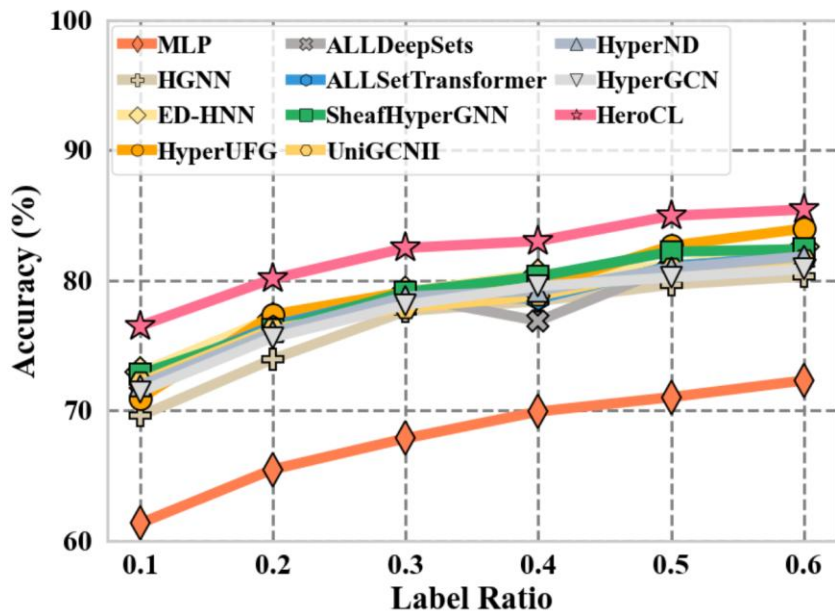
□ Comparison on heterophilic datasets

Model	Actor (0.47)	Amazon (0.37)	Twitch (0.49)	Pokec (0.45)	Senate (0.46)	House (0.49)	Ave. Rank
MLP	85.45 ± 1.21	26.70 ± 2.82	52.77 ± 1.81	56.92 ± 2.46	52.25 ± 5.17	51.86 ± 2.34	5.17
HGNN	74.47 ± 0.32	23.79 ± 0.24	51.88 ± 0.26	49.82 ± 0.27	48.59 ± 4.52	61.39 ± 2.96	8.33
HyperGCN	68.67 ± 4.38	22.53 ± 3.94	51.32 ± 1.02	52.43 ± 3.68	42.45 ± 3.67	48.32 ± 2.93	9.50
UniGCNII	80.48 ± 1.13	26.63 ± 1.32	50.84 ± 0.76	54.25 ± 2.70	49.30 ± 4.25	67.25 ± 2.57	7.50
HyperND	92.52 ± 0.81	26.08 ± 0.33	51.44 ± 0.67	55.94 ± 0.45	52.82 ± 3.20	51.70 ± 3.37	5.67
AllDeepSets	82.00 ± 2.33	18.60 ± 0.17	50.72 ± 0.96	51.11 ± 1.04	48.17 ± 5.67	67.82 ± 2.40	8.92
AllSetTransformer	83.39 ± 1.73	18.60 ± 0.17	50.45 ± 0.76	58.40 ± 0.42	51.83 ± 5.22	69.33 ± 2.20	7.25
ED-HNN	91.86 ± 0.43	26.21 ± 0.36	50.86 ± 0.88	59.11 ± 0.57	64.79 ± 5.14	72.45 ± 2.28	4.50
SheafHyperGNN	80.09 ± 2.45	26.93 ± 3.04	51.03 ± 0.76	55.34 ± 4.39	68.73 ± 4.68	73.84 ± 2.30	5.00
HyperUFG	89.32 ± 0.75	40.53 ± 2.25	52.35 ± 0.04	62.30 ± 0.12	67.61 ± 7.00	72.82 ± 2.22	2.33
HeroCL (Ours)	85.58 ± 0.23	30.81 ± 0.28	52.82 ± 0.12	59.83 ± 0.28	75.35 ± 4.46	77.28 ± 3.10	1.83

Experiments

Robustness under Varying Label Ratios

- On the homophilic dataset, all methods degrade smoothly as supervision decreases
- On the heterophilic dataset, all methods suffer severe performance drops



□ Ablation Study

Model Variant	Cora-CA	Citeseer	Senate	House
w/o LabelCL	83.18 ± 1.42	72.97 ± 0.50	75.28 ± 4.47	76.52 ± 1.20
w/o StructCL	86.54 ± 1.33	75.18 ± 0.43	75.30 ± 4.63	76.57 ± 1.28
w/o LS-CL	83.24 ± 1.67	73.37 ± 0.33	75.18 ± 4.61	76.48 ± 1.19
w/o MHNE	85.37 ± 0.75	74.29 ± 0.36	73.82 ± 4.12	76.64 ± 1.19
Full	86.37 ± 1.08	74.96 ± 0.35	75.35 ± 4.46	77.28 ± 3.10

Conclusion

- **HNNs show limited performance due to two key factors**
 - Label dependency
 - Structural sparsity

- **HGCL mitigates these bottlenecks through contrastive self-supervision**
 - Pull positive pair closer, push negative pair away

- **Effective HGCL requires two key design choices**
 - How to augment a hypergraph?
 - What to contrast?